

KYTRON SOLUTION FOR INJECTION/ INFUSION 3 MG/ 3 ML

DESCRIPTION:

KYTRON SOLUTION FOR INJECTION/INFUSION 3MG/3ML:

Un-opened Ampoule: A clear, colorless solution.

After Dilution: A clear, colorless solution after the dilution with compatible diluents.

COMPOSITION:

Kytron Solution for Injection/Infusion 3mg/3ml is available as an ampoule containing granisetron hydrochloride, equivalent to 3mg of granisetron in 3ml.

PHARMACODYNAMICS:

Granisetron HCL is a potent antiemetic and highly selective antagonist of 5-hydroxytryptamine (5-HT) 3 receptors. Antagonism of 5-HT₃ receptors located peripherally on vagal nerve terminals and centrally in the chemoreceptor trigger zone in the area postrema, is one of the most effective pharmacological methods of preventing cytotoxic-induced emesis. Mucosal enterochromaffin cells release serotonin during chemotherapy-induced emesis. Serotonin stimulates 5-HT₃ receptors and evokes a vagal afferent discharge to subsequently induce emesis.

PHARMACOKINETICS:

Granisetron is rapidly absorbed after oral doses, with peak plasma concentrations occurring after about 2 hours. Oral bioavailability is about 60% as a result of first-pass hepatic metabolism. Granisetron has a large volume of distribution of around 3 litres/ kg; plasma protein binding is about 65%. The pharmacokinetics exhibit considerable interindividual variation and the elimination half-life after an intravenous dose is reported to be around 4 to 5 hours in healthy subjects but about 9 to 12 hours in cancer patients. It is metabolised in the liver, primarily by N-demethylation, with less than 20% of a dose recovered unchanged in urine, the remainder being excreted in faeces and urine as metabolites. Granisetron clearance is not affected by renal impairment, but is lower in the elderly and in patients with hepatic impairment.

INDICATIONS:

Kytron Solution for Injection/Infusion 3mg/3ml is indicated for the prevention and treatment (control) of a) acute and delayed nausea and vomiting associated with chemotherapy and radiotherapy b) post-operative nausea and vomiting

RECOMMENDED DOSAGE:

Standard Dosage Chemotherapy Induced Nausea and Vomiting (CINV)

Adults Prevention:

A dose of 1-3 mg (10-40mcg/kg) of Kytron Solution for Injection/Infusion 3mg/3ml should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes, prior to the start of chemotherapy.

Treatment: A dose of 1-3 mg (10-40mcg/kg) Kytron Solution for Injection/Infusion 3mg/3ml should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes. Further treatment doses of Kytron Solution for Injection/Infusion 3mg/3ml may be administered, if required, at least 10 minutes apart. The maximum dose Kytron Solution for Injection/Infusion 3mg/3ml to be administered over 24 hours should not exceed 9 mg.

Pediatrics Prevention and treatment:

A dose of 10-40 mcg/kg body weight (up to 3 mg) should be administered as an intravenous infusion, diluted in 10 to 30 ml infusion fluid and administered over 5 minutes prior to the start of chemotherapy. One additional dose may be administered within a 24 hour period if required. This additional dose should not be administered until at least 10 minutes after the initial infusion.

Radiotherapy Induced Nausea and Vomiting (RINV)

Adults Prevention:

A dose of 1-3 mg (10-40mcg/kg) of Kytron Solution for Injection/Infusion 3mg/3ml should be administered either as a slow intravenous injection (over 30 seconds) or as an intravenous infusion diluted in 20 to 50 ml infusion fluid and administered over 5 minutes, prior to the start of radiotherapy.

Treatment: There is insufficient information to recommend the intravenous administration of Kytron Solution for Injection/Infusion 3mg/3ml in the treatment of RINV in adult patients.

Pediatrics:

There is insufficient information to recommend intravenous administration of Kytron Solution for Injection/Infusion 3mg/3ml in the prevention and treatment of RINV in children.

Post-operative Nausea and Vomiting (PONV)

Adults Prevention: A dose of 1 mg (10mcg/kg) of Kytron Solution for Injection/Infusion 3mg/3ml should be administered as a slow intravenous injection (over 30 seconds) prior to induction of anesthesia.

Treatment: A dose of 1 mg (10mcg/kg) of Kytron Solution for Injection/Infusion 3mg/3ml should be administered by slow intravenous injection (over 30 seconds). The maximum dose for patients undergoing anesthesia for surgery is a total dose of 3 mg of Kytron Solution for Injection/Infusion 3mg/3ml i.v. in one day.

Pediatrics:

There is insufficient information to recommend intravenous administration of Kytron Solution for Injection/Infusion 3mg/3ml in the prevention and treatment of postoperative nausea and vomiting in pediatric patients.

Special Dosage Instructions Geriatrics: No dosage adjustments required. Renal impairment: No dosage adjustments required. Hepatic Impairment: No dosage adjustments required.

Preparing the Solution with Compatible Diluent

Adults: To prepare the dose of 3 mg, withdraw 3 ml from the ampoule and dilute with a compatible infusion fluid to a total volume of 20 to 50 ml, in any of the following solutions: 0.9% sodium chloride, 0.18% sodium chloride and 4% glucose, 5% glucose, Hartmann's solution and mannitol.

Paediatric patients: To prepare the dose of 40 µg/kg, the appropriate volume is withdrawn (up to 3 ml from the ampoule) and diluted with infusion fluid (as for adults) to a total volume of 10 to 30 ml.

The injectable presentations contain no antimicrobial agent. Use once and discard any residue.

In order to reduce microbiological hazards it is recommended that the prepared infusion be commenced as soon as practicable after its preparation and should be completed within 24 hours. If not used immediately, in use storage times and conditions prior to use are the responsibility of the user and would normally not be longer than 24 hours at 25°C.

As a general precaution, Kytron Solution for Injection/Infusion 3mg/3ml should not be mixed in solution with other drugs other than dexamethasone sodium phosphate.

Admixtures of granisetron hydrochloride and dexamethasone sodium phosphate are compatible at concentrations of 10 to 60 µg/mL granisetron and 80 to 480 µg/mL dexamethasone phosphate in either 0.9% sodium chloride or 5% glucose intravenous infusion fluids. Stability data have been provided to demonstrate the physical and chemical stability of these solutions when stored at 25°C exposed to light for up to 24 hours. To reduce microbiological contamination hazards, it is recommended that admixing should be effected immediately prior to use and infusion commenced as soon as practicable after preparation. The admixture will have a shelf-life of 24 hours.

Parenteral drug products should be inspected visually for particulate matter and discolouration before administration whenever solution and container permit.

ROUTE OF ADMINISTRATION : For IV Injection and IV Infusion

CONTRAINDICATIONS:

Kytron Solution for Injection/Infusion 3mg/3ml is contraindicated in patients with known hypersensitivity to granisetron or to any of its excipients.

WARNINGS AND PRECAUTIONS:

As Kytron Solution for Injection/Infusion 3mg/3ml may reduce lower bowel motility, patients with signs of sub-acute intestinal obstruction should be monitored following administration of Kytron Solution for Injection/Infusion 3mg/3ml. Therefore, caution should be exercised in patients with cardiac co-morbidities, on cardio-toxic chemotherapy and/or with concomitant electrolyte abnormalities. There are no data on the effect on Kytron Solution for Injection/Infusion 3mg/3ml on the ability to drive, however there have been occasional reports of somnolence in clinical studies which should be taken into account. No special precautions are required for the elderly or patients with renal or hepatic impairment. Cross-sensitivity between 5-HT₃ antagonists has been reported. As with other 5-HT₃ antagonists, cases of serotonin syndrome (including altered mental status, autonomic dysfunction and neuromuscular abnormalities) have been reported following the concomitant use of Kytron Solution for Injection/Infusion 3mg/3ml and other serotonergic drugs. If concomitant treatment with granisetron and other serotonergic drugs is clinically warranted, appropriate observation of the patient is advised. As for other 5-HT₃ antagonists, cases of ECG modifications including QT prolongation have been reported with Kytron Solution for Injection/Infusion 3mg/3ml. The ECG changes with Kytron Solution for Injection/Infusion

3mg/3ml were minor, generally not of clinical significance and specifically, there was no evidence of proarrhythmia. However, in patients concurrently treated with drugs, known to prolong QT interval and/or are arrhythmogenic, this may lead to clinical consequences.

INTERACTIONS WITH OTHERS MEDICAMENTS:

The metabolism of Granisetron is induced by phenobarbital

Avoid concomitant use of Granisetron with Apomorphine as Granisetron may increase Apomorphine levels/effect.

Kytron Solution for Injection/Infusion 3mg/3ml has been safely administered with benzodiazepines, neuroleptics, and anti-ulcer medications commonly prescribed with anti-emetic treatments. Additionally, Kytron Solution for Injection/Infusion has shown no apparent drug interaction with emetogenic cancer chemotherapies. As for other 5-HT₃ antagonists, cases of ECG modifications including QT prolongation have been reported with Kytron Solution for Injection/Infusion. The ECG changes with Kytron Solution for Injection/Infusion were minor, generally not of clinical significance and specifically, there was no evidence of proarrhythmia. However, in patients concurrently treated with drugs known to prolong QT interval and/or are arrhythmogenic, this may lead to clinical consequences. As with other 5-HT₃ antagonists, cases of serotonin syndrome have been reported following the concomitant use of Kytron Solution for Injection/Infusion and other serotonergic drugs. If concomitant treatment with granisetron and other serotonergic drugs is clinically warranted, appropriate observation of the patient is advised.

Kytron Solution for Injection/Infusion 3mg/3ml cannot be diluted with other diluent except stated in 'Preparing the Solution with Compatible Diluent'

USE IN PREGNANCY AND LACTATION:

Use in Pregnancy:

There are no adequate or well controlled studies of Granisetron in human pregnancy. Teratogenic effects were not observed in animal studies. Because of the low safety margin indicated by the animal studies and because animal reproduction studies are not always predictive of human response, Kytron Solution for Injection/Infusion 3mg/3ml should be used during pregnancy only if clearly needed.

Use in Lactation:

There are no data on the excretion of Granisetron in human breastmilk, therefore use of the drug during lactation should be limited to situations where the potential benefit to the mother justifies the potential risk to the nursing infant.

SIDE EFFECTS:

Gastrointestinal disorders: Common: Constipation.

Central Nervous system: Common: Headache, agitation, anxiety, CNS stimulation, dizziness, insomnia, somnolence; Rare: extrapyramidal syndrome (only in presence of other drugs associated with this syndrome). Uncommon: Serotonin syndrome

Body as a whole: Common: fever.

Cardiovascular: Common: hypertension; Rare: hypotension, arrhythmias, sinus bradycardia, atrial fibrillation, varying degrees of A-V block, ventricular ectopy including non-sustained tachycardia, ECG abnormalities, angina pectoris, syncope, QT prolongation

Hypersensitivity: Rare: hypersensitivity reactions (e.g. anaphylaxis, shortness of breath, hypotension, urticaria).

Hepatic: Common: transient increases in AST and ALT. These are generally within the normal range and have been reported at similar frequency in patients receiving comparator therapy.

Dermatological: Common: skin rashes.

Special Senses: Common: taste disorder.

Other common events often associated with chemotherapy also have been reported: leucopenia, decreased appetite, anaemia, alopecia, and thrombocytopenia. As with other 5-HT₃ antagonists, cases of serotonin syndrome (including altered mental status, autonomic dysfunction and neuromuscular abnormalities) have been reported following the concomitant use of Kytron Solution for Injection/Infusion 3mg/3ml and other serotonergic drugs.

SYMPTOMS AND TREATMENT OF OVERDOSE:

There is no specific antidote for Kytron Solution for Injection/Infusion 3mg/3ml. In the case of over dosage, symptomatic treatment should be given.

INCOMPATIBILITIES: Not applicable.

STORAGE CONDITIONS:

Item		Storage Condition	Other storage instructions
Unopened ampoule:		Store at room temperature below 30°C and protect from light.	Keep out of reach of children. <i>Jauhkan daripada kanak-kanak.</i>
After dilution:	Compatible Diluents: 0.9% Sodium Chloride, 0.18% Sodium Chloride & 4% Glucose, 5% Glucose, Hartmann's solution, Mannitol	The admixtures solutions are would normally stable not be longer than 24 hours at 25°C.	
	Compatible Drugs Admixtures: Dexamethasone Sodium Phosphate, 0.9% Sodium Chloride, 5% Glucose	The admixtures solutions are stable when stored at 25°C exposed to light for up to 24 hours.	

PACK SIZES: Pack in 10's ampoules of 3 ml / box.

SHELF LIFE: Please refer to outer package.

PRODUCT REGISTRATION HOLDER AND MANUFACTURER:

DUOPHARMA (M) SDN BHD

Lot 2599, Jalan Seruling 59, Kawasan 3,

Taman Klang Jaya, 41200 Klang, Selangor, MALAYSIA